Orthogonal and orthonormal sets

All vectors lie in $\mathbb{R}^n$, with the usual dot product and norm $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$.

Dot product (inner product): $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T\mathbf{v} = \langle \mathbf{u}, \mathbf{v} \rangle$ for column vectors.

Orthogonal vectors: $\mathbf{x} \cdot \mathbf{y} = 0$.

Orthogonal set: $\{\mathbf{f}_1, \dots, \mathbf{f}_k\}$ with $\mathbf{f}_i \neq \mathbf{0}$ for every i, and $\mathbf{f}_i \cdot \mathbf{f}_j = 0$ whenever $i \neq j$. The zero vector is excluded from an orthogonal set.

Orthonormal set: an orthogonal set in which every vector has norm 1; equivalently, $\mathbf{q}_i \cdot \mathbf{q}_j = 1$ if $i = j$ and 0 otherwise.

Orthogonal / orthonormal basis: a basis of a subspace that is an orthogonal / orthonormal set, respectively.

Consequences of orthogonality

Pythagoras' theorem: for an orthogonal set $\{\mathbf{f}_1, \dots, \mathbf{f}_k\}$,

$$\|\mathbf{f}_1 + \cdots + \mathbf{f}_k\|^2 = \|\mathbf{f}_1\|^2 + \cdots + \|\mathbf{f}_k\|^2.$$

Linear independence: every orthogonal set is linearly independent, and hence is a basis of its span.

Normalization: given an orthogonal set $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, set

$$\mathbf{q}_i = \frac{\mathbf{f}_i}{\|\mathbf{f}_i\|}, \qquad 1 \leq i \leq m.$$

The vectors $\mathbf{q}_i$ form an orthonormal basis of the same span. Normalization requires $\mathbf{f}_i \neq \mathbf{0}$.

Expansion and coordinates

Expansion theorem: let $\{\mathbf{q}_1, \dots, \mathbf{q}_m\}$ be an orthonormal basis of a subspace $U \subseteq \mathbb{R}^n$. For every $\mathbf{x} \in U$,

$$\mathbf{x} = \sum_{i=1}^{m} (\mathbf{x} \cdot \mathbf{q}_i)\mathbf{q}_i.$$

The coefficient of $\mathbf{q}_i$ is the dot product $\mathbf{x} \cdot \mathbf{q}_i$.

Hypotheses matter: the vector must lie in U, and the basis must be orthonormal. For an orthogonal basis, normalize first before using this formula.

Coordinate vector: for the ordered orthonormal basis $B = (\mathbf{q}_1, \dots, \mathbf{q}_m)$ and $\mathbf{u} \in U$,

$$C_B(\mathbf{u}) = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix} = \begin{bmatrix} \mathbf{u} \cdot \mathbf{q}_1 \\ \vdots \\ \mathbf{u} \cdot \mathbf{q}_m \end{bmatrix}.$$

The entry order follows the chosen order of the basis vectors.